Data Structures and Algorithms

1. Arrays

An array is a data structure that stores elements in contiguous memory locations. Elements can be accessed using an index. Accessing an element by index takes O(1) time.

```
Example:
arr = [10, 20, 30, 40]
print(arr[2])  # 30
```

2. Binary Search

Binary search is an efficient searching algorithm for a sorted array. It repeatedly divides the search range into two halves. The time complexity of binary search is O(log n).

```
Example:
Sorted array = [10, 20, 30, 40, 50]
Target = 40
The search repeatedly checks the middle element.
```

3. Stack

A stack follows the LIFO (Last In, First Out) principle. The last element inserted is the first element removed.

```
Example:
stack = []
stack.append(10)
stack.append(20)
stack.pop()  # removes 20
```

4. Queue

A queue follows the FIFO (First In, First Out) principle. The first element inserted is the first element removed.

5. Linked List

A linked list is a collection of nodes where each node stores data and a reference to the next node. Inserting or deleting a node can be efficient when its position is known.